

Lines & Angles — Practice Questions

Digital SAT Style | Mixed Difficulty | Total Questions: 25

Q1.

DIFFICULTY: EASY

In the figure, two straight lines intersect at a point. One of the four angles formed measures $(3x + 12)^\circ$. The angle vertically opposite to it measures $(5x - 24)^\circ$. What is the value of x ?

- A) 15
 - B) 18
 - C) 20
 - D) 24
-

Q2.

DIFFICULTY: EASY

Two lines intersect forming four angles. One angle measures 74° . What is the sum of the two angles that are NOT vertically opposite to the 74° angle and NOT equal to 74° ?

- A) 74°
 - B) 106°
 - C) 212°
 - D) 148°
-

Q3.

DIFFICULTY: EASY

Line m and line n are parallel. A transversal crosses both lines. The co-interior (same-side interior) angles are $(4k - 10)^\circ$ and $(2k + 40)^\circ$. What is the value of k ?

- A) 20
 - B) 22
 - C) 25
 - D) 30
-

Q4.

DIFFICULTY: EASY

An angle and its supplement are in the ratio 2 : 7. What is the measure of the smaller angle?

- A) 20°
 - B) 40°
 - C) 60°
 - D) 140°
-

Q5.

DIFFICULTY: EASY

Points A, B, and C are collinear with B between A and C. Ray BD bisects angle ABE, where E is not on line AC. If angle ABE = 64° , what is the measure of angle DBC?

- A) 32°
 - B) 116°
 - C) 148°
 - D) 158°
-

Q6.

DIFFICULTY: MEDIUM

Lines p and q are parallel, cut by transversal t. The alternate interior angles are $(7m - 9)^\circ$ and $(4m + 18)^\circ$. What is the measure of each angle?

- A) 49°
 - B) 54°
 - C) 58°
 - D) 63°
-

Q7.

DIFFICULTY: MEDIUM

Two parallel lines are cut by a transversal. One angle at the first intersection measures $(6x + 5)^\circ$ and the angle in the corresponding position at the second intersection measures $(3x + 20)^\circ$. What is their relationship?

- A) Alternate interior; equal.
 - B) Co-interior; supplementary.
 - C) Corresponding; equal.
 - D) Complementary; sum to 90° .
-

Q8.

DIFFICULTY: MEDIUM

Line r is parallel to line s. A transversal forms alternate exterior angles: $(9a - 7)^\circ$ at line r and $(5a + 33)^\circ$ at line s. What is the value of a?

- A) 8
 - B) 10
 - C) 12
 - D) 14
-

Q9.

DIFFICULTY: MEDIUM

An angle measures $(2y + 15)^\circ$. Its complement measures $(3y - 20)^\circ$. What is the value of y?

- A) 15
- B) 19

- C) 21
 - D) 24
-

Q10.

DIFFICULTY: MEDIUM

Three lines meet at a point. Three consecutive angles on one side of a straight line are $(2x + 10)^\circ$, $(x + 20)^\circ$, and $(x - 10)^\circ$. What is the value of x ?

- A) 30
 - B) 35
 - C) 40
 - D) 45
-

Q11.

DIFFICULTY: MEDIUM

Lines $AB \parallel CD$. Transversal EF creates alternate interior angles: angle $AEF = (5t + 8)^\circ$ and angle $EFD = (3t + 40)^\circ$. What is the measure of angle AEF ?

- A) 82°
 - B) 88°
 - C) 93°
 - D) 98°
-

Q12.

DIFFICULTY: MEDIUM

A ray divides an angle into two parts. The larger part is 24° more than three times the smaller part. The original angle is 100° . What is the smaller part?

- A) 19°
 - B) 22°
 - C) 25°
 - D) 28°
-

Q13.

DIFFICULTY: MEDIUM

Two parallel lines are cut by a transversal creating a 55° angle at the first line. What is the co-interior angle on the same side at the second line?

- A) 55°
 - B) 115°
 - C) 125°
 - D) 135°
-

Q14.

DIFFICULTY: MEDIUM

Points P, Q, R are collinear. Ray QS makes angle $PQS = (6n + 4)^\circ$ and angle $SQR = (3n - 4)^\circ$. Ray QT bisects angle SQR. What is angle SQT?

- A) 24°
 - B) 28°
 - C) 32°
 - D) 36°
-

Q15.

DIFFICULTY: HARD

Line j $\parallel$ line k. A transversal creates a 63° angle at line j. The co-interior angle at line k is $(4p + 7)^\circ$. What is p?

- A) 23.5
 - B) 25
 - C) 27.5
 - D) 30
-

Q16.

DIFFICULTY: HARD

Two parallel streets are crossed by a diagonal road. Alternate exterior angles are $(5x + 15)^\circ$ and $(9x - 25)^\circ$. What is the angle at the first street?

- A) 55°
 - B) 60°
 - C) 65°
 - D) 70°
-

Q17.

DIFFICULTY: HARD

Three lines form a triangle with interior angles $(2x)^\circ$, $(3x - 10)^\circ$, and $(x + 22)^\circ$. What is the exterior angle at the $(2x)^\circ$ vertex?

- A) 118°
 - B) 124°
 - C) 128°
 - D) 136°
-

Q18.

DIFFICULTY: HARD

AB $\parallel$ CD. Transversal EG creates co-interior angles $(5q - 10)^\circ$ at E and $(3q + 30)^\circ$ at F. What is q?

- A) 16
- B) 18
- C) 20

D) 24

Q19.

DIFFICULTY: HARD

Two rays share an endpoint forming an angle of $(5x + 3x^2)^\circ$ for $x = 3$. An adjacent angle forms a straight angle. A ray bisects the second angle. What is each half?

- A) 57°
 - B) 64°
 - C) 69°
 - D) 75°
-

Q20.

DIFFICULTY: HARD

Lines $m \parallel n$. Co-interior angles are $(2a + b)^\circ$ and $(a + 2b)^\circ$. Given $a - b = 6$, what is $a + b$?

- A) 54
 - B) 60
 - C) 66
 - D) 72
-

Q21.

DIFFICULTY: HARD

Segment PQ has midpoint M. Ray $MR \perp PQ$. Ray MS bisects angle RMQ. What is angle PMS?

- A) 90°
 - B) 120°
 - C) 135°
 - D) 150°
-

Q22.

DIFFICULTY: HARD

Three lines meet at a point forming six angles. Three consecutive angles on one side of a line are $(2x + 5)^\circ$, $(x + 15)^\circ$, and $(x + 20)^\circ$. What is x ?

- A) 30
 - B) 35
 - C) 40
 - D) 45
-

Q23.

DIFFICULTY: HARD

Parallel lines a and b are cut by transversal c . Co-interior angles are $(12k - 6)^\circ$ at line a and $(6k + 42)^\circ$ at line b . What is the angle at line a ?

- A) 78°
 - B) 84°
 - C) 90°
 - D) 96°
-

Q24.

DIFFICULTY: HARD

Line ℓ passes through the origin with slope 1. Line m is horizontal (slope 0). What is the acute angle between ℓ and m ?

- A) 30°
 - B) 45°
 - C) 60°
 - D) 90°
-

Q25.

DIFFICULTY: HARD

Two parallel lines are cut by two transversals. Each transversal makes a 42° angle with the parallel lines. A student claims the transversals are parallel. Which is correct?

- A) Correct: equal corresponding angles means the transversals are parallel.
 - B) Incorrect: the transversals make supplementary angles.
 - C) Incorrect: equal angles only apply for one transversal, not between two.
 - D) Incorrect: the transversals are perpendicular because $42^\circ + 42^\circ \approx 90^\circ$.
-